
NPU instructions

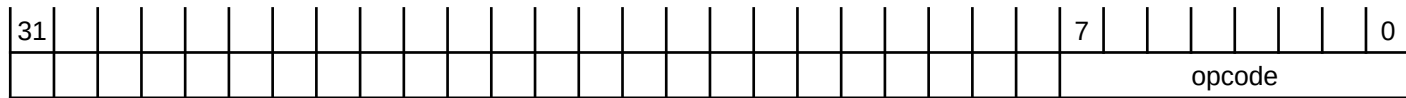
Mar. 8, 2026

Park, Hyung chul

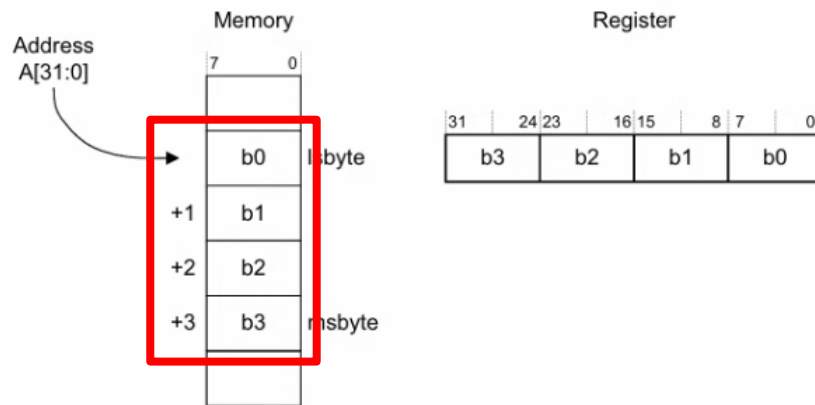
NPU instructions

Instruction format

- Instruction length : 32 bits



- Little-endian



SIMD instructions

Instructions	Opcode	Memo
Vector add	b0000_0001	{vector, vector}, {vector, scalar}, {vector, immediate}
Vector subtract	b0000_0010	{vector, vector}, {vector, scalar}, {vector, immediate}
Vector bitwise logical	b0000_1000	{vector, vector}, {vector, scalar}, {vector, immediate}
Vector shift	b0000_1001	{vector, vector}, {vector, scalar}, {vector, immediate}
Vector multiply	b0000_1010	{vector, vector}, {vector, scalar} , {vector, immediate}
Vector divide	b0000_1011	{vector, vector}, {vector, scalar} , {vector, immediate}
Vector multiply-add	b0000_1100	{vector, vector}, {vector, scalar} , {vector, immediate}
Vector move	b0000_1101	{vector, vector}, {vector, scalar} , {vector, immediate}
Vector square-root	b0000_1110	
Vector exponential	b0000_1111	
Vector compare	b0001_0001	{vector, vector}, {vector, scalar}, {vector, immediate}
Vector Min/Max	b0001_0010	{vector, vector}, {vector, scalar}
Type-convert	b0001_0011	Float $\leftrightarrow$ integer
Reduce sum	b0001_0100	vector
Broadcast	b0001_0101	Immediate, scalar
Sign inversion	b0001_0110	vector
Vector copy	b0001_0111	vector
Cos / sin	B0001_1000	vector

Matrix instructions

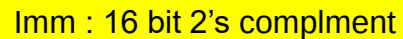
Instructions	Opcode	Memo
Matrix add	b0100_0000	{matrix, matrix}, {matrix, scalar}, {matrix, immediate}
Matrix subtract	b0100_0001	{matrix, matrix}, {matrix, scalar}, {matrix, immediate}
Matrix multiply	b0100_0010	{matrix, matrix}, {matrix, scalar}, {matrix, immediate}
Matrix move	b0100_0011	{matrix, matrix}, {matrix, scalar}, {matrix, immediate}

Data (Memory) instructions

Instructions	Opcode	Memo
Set {Vector, Matrix } start address	b1000_0000	Global buffer start address (low) of {Vector, Matrix} for load, save
Set Vector length	b1000_0010	Global buffer length, Vector, data
Set Matrix tile	b1000_1000	Set matrix dimension
Load {vector, matrix}	b1001_0000	
Save result	b1001_1000	

- Instructions

- { vector, immediate } (ex : vector – immediate)



– { vector, vector_2 } (ex : vector – vector_2)



- Vector bitwise logical,
 - { vector, immediate } (ex : vector & immediate)

Imm : unsigned 16 bit

- | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|----|----|----------|----|----|--|--|--|----------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------------|--|---|---|--|--|--|--|--|--|--|---|
| 31 | 30 | 29 | 28 | 27 | | | | 23 | | | | | | | | | | | | | | | | | | 8 | 7 | | | | | | | | 0 |
| 0 | 1 | operator | | | | | | Buffer address | | | | | | | | | | | | | | | | b0000_1000 | | | | | | | | | | | |

- [illegible]

- 3/7

Instructions - detail

- Instructions

broadcast			

- {immediate }

31	30							23												8	7							0
0	0							Imm[15:0]																				

Imm : 16 bit 2's complement

- {scalar }

31	30							23												8	7							0
0	1							Buffer address																				

Instructions - detail

- Vector cos/sin,

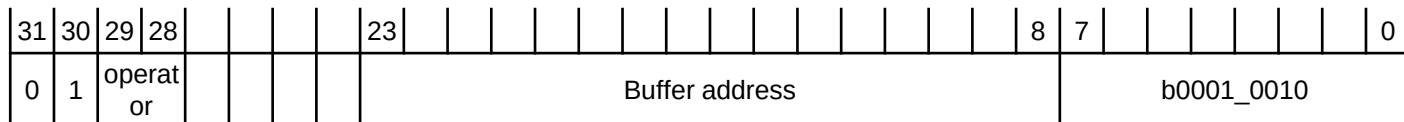
[illegible]

- Operator
 - '000' : cos
 - '001' : sin

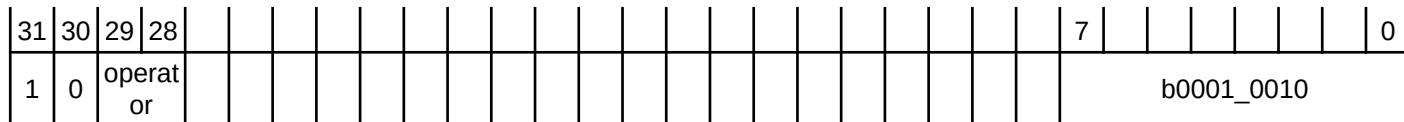
Instructions - detail

- Vector min/max,

- { vector, scalar }



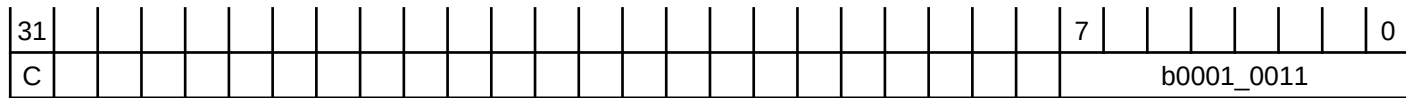
- { vector, vector } (ex : vector & vector_2)



- Operator

- '00' : min
 - '01' : max

- Type-convert



- C : 'control'
 - 0 : integer $\rightarrow$ float
 - 1 : float $\rightarrow$ integer

- Instructions

- { vector, immediate } (ex : vector – immediate)

- { vector, scalar }

- { vector, vector_2 } (ex : vector - vector_2)

- a : 'activation function'

- 0 : OFF
- 1 : ON

- b : MAC (multiply-accumulate) operation
 - 0 : OFF
 - 1 : ON

Instructions - detail

- Instructions

Matrix multiply			
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- { vector, immediate } (ex : vector – immediate)

31	30	29	28					23										8	7						0
0	0	a	b					Imm[15:0]										b0100_0010							

Imm : 16 bit 2's complement

- { vector, scalar }

31	30	29	28					23										8	7						0
0	1	a	b					Buffer address										b0100_0010							

- { vector, vector_2 } (ex : vector – vector_2)

31	30	29	28																7						0
1	0	a	b																b0100_0010						

- a : 'activation function'

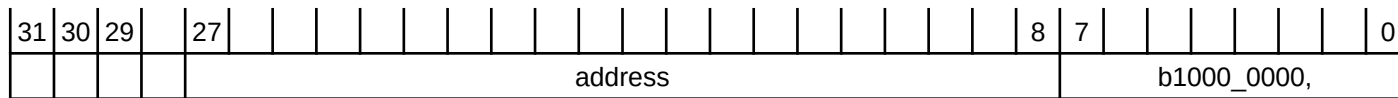
- 0 : OFF
- 1 : ON

- b : MAC (multiply-accumulate) operation

- 0 : OFF
- 1 : ON

Instructions - detail

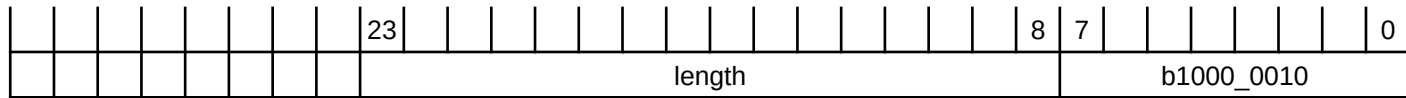
- Set {Vector, Matrix } start address



- b [31:30]
 - '00' : load address of 1st {vector, matrix}
 - '01' : load address of 2nd {vector, matrix}
 - '10' : save address
- b [29]
 - 0 : low address part
 - 1 : high address part

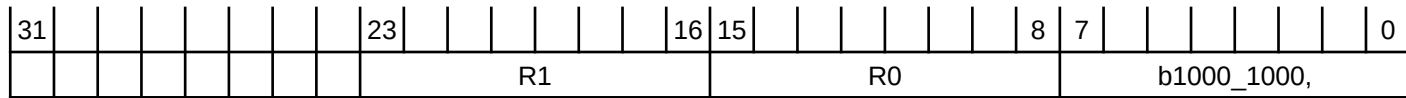
Instructions - detail

- Set Vector length

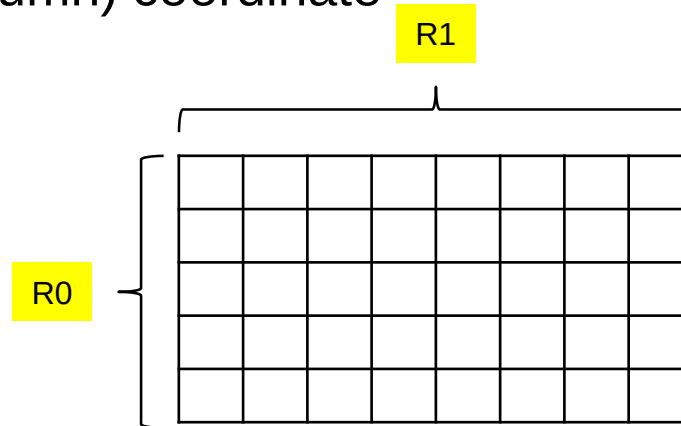


Instructions - detail

- Matrix set tile

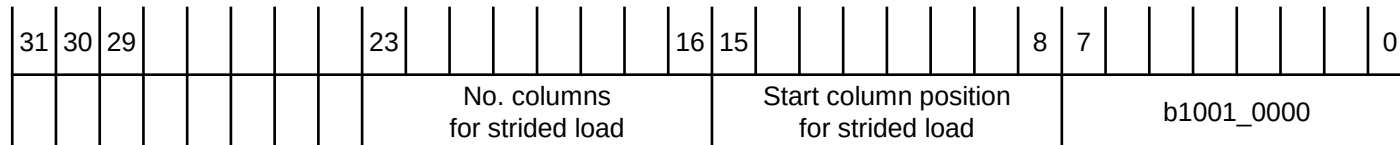


- [31]
 - 0 : 1st matrix
 - 1 : 2nd matrix
- R0 : 1st (row) coordinate
- R1 : 2nd (column) coordinate



Instructions - detail

- Load {vector, matrix}



– [31]

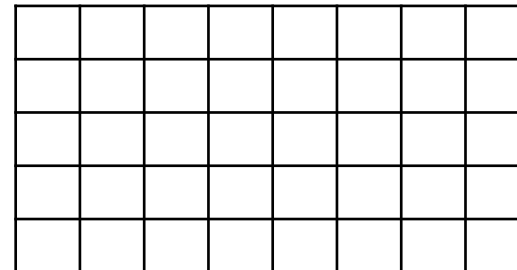
- 0 : vector
- 1 : matrix

– [30]

- 0 : 1st
- 1 : 2nd

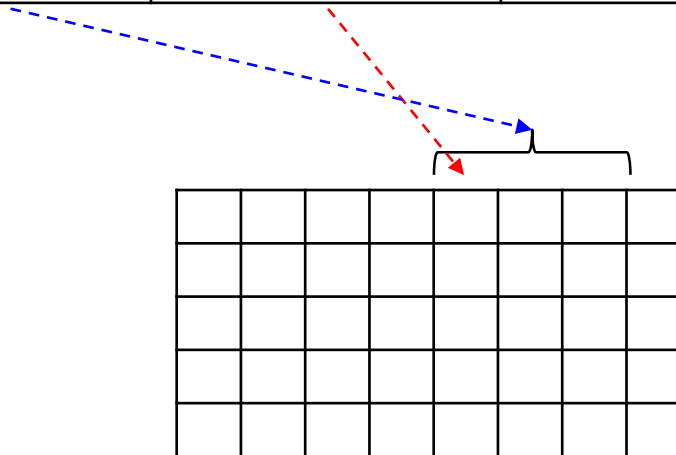
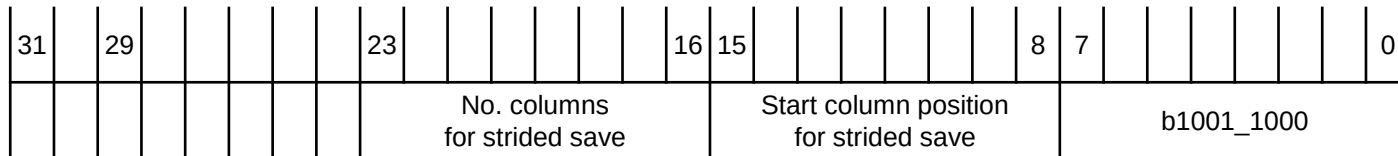
– [29] : strided (column vector) load (bit [31] : 1)

- 0 : disable
- 1 : enable



Instructions - detail

- Save result



- [31]
 - 0 : vector
 - 1 : matrix
- [29] : strided (column vector) save (bit [31] : 1)
 - 0 : disable
 - 1 : enable